

Weak signal detection and identification in large data sets: a review of methods and applications

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Abstract

The detection of anomalies, or event forecast, relies on understanding past and present events. Evaluating the dynamics of the received signals and determining which ones may have a relevant future is a crucial task. It is an important task for different levels of data treatment, from a simple log analysis tool to the definition of the company's strategic planning for the next few years. In recent years, there has been an increasing demand for anticipating emerging issues or so-called weak signals. However, as these signals are small precursors, usually is lost in the middle of a large amount of noisy information, determining what is essential and what is not remains a complex problem. Different researchers have applied weak signal detection in a broad set of fields, using distinct methods. This survey provides a comprehensive and ordered state-of-the-art review of the weak signals, early warnings and detection field. We focus on methods and techniques that could be applied to a broad range of domains. We also propose a taxonomy of the methods and discuss and compare the characteristics of different approaches. This survey presents a theoretical background of the field and some of the most interesting applications in the domain of weak signal detection.

Keywords: weak signal, early warning, event detection, anomaly detection

1 Introduction

Data mining is the process of finding anomalies, patterns and correlations within colossal data sets. This process supports decision-making by analyzing and investigating historical data in order to predict events and outcomes.

We witness a tremendous increase in the volume and diversity of available information, so much that we are sometimes unable to see the small significant clues, that can give us a warning of important events to come. Finding these clues is even harder if the events we are interested in are not known beforehand. In this study, an event can be considered as any fact, information, or even a series of data that can be individually identified, isolated and have a meaning and a perceived impact. We consider a "weak signal" any piece of information that can be identified as a precursor warning of something that subsequently will turn into an event. Identifying such signals, especially assuming no prior knowledge, is a challenging and interesting task. The concept of early warning signals is used in organizations and companies to adapt to emerging changes. Based on the recognition of changes, this continually evolving process requires a global vision of current circumstances and identifying the factors leading to profound changes.

The field of weak signal identification has applications in various and diverse areas, like for example, weak signals related to identifying terrorism or mass transport attacks [1]; strategic decision [2]; predicting tipping points from the London riots in 2011 [3]; predictive early biomarkers [4]. In this review, we collected the different methods to detect and isolate weak signals over diverse data sets, regardless of the application area, focusing on reproducible methods and not extremely specific to a use case.

Note that in this paper, we decide to focus on the work of weak signal detection in Big Data analysis and not in sense of signal processing such as stochastic resonance.

1.1 Background

[5] describes a weak signal as a "sudden, urgent, unfamiliar changes in the firm's perspective which threaten either a major profit reversal or loss of a major opportunity" and considers that weak signals detection allows for better decision-making and foresight planning. Ansoff, the weak signal concept's precursor, considers a weak signal as the "warning (external or internal), events and developments that are still too incomplete to allow for an accurate estimate of their impact and/or to determine a full adapted response". However, even if with strong roots and intuitively sound, the weak signal concept is not accurately defined in the literature. A weak signal is both an event announcer

and an informal alert, even if it does not point to exact information. This provides some information about the future but its impact cannot be quantified accurately, as its estimation relies upon the actions that follow the signal [6]. The interpretation of such a signal is challenging as it is hidden under the rest of the data. Scanning for and using them in scenario work is considered a successful way to look towards the future.

However, the definition of the weak signal poses a problem, as various authors define the concept differently. Different authors use different terms to reference weak signals, [7] employs "Future signs", [8] "emerging issues", [9] "seeds of changes", "fact emerging" [10], and [11] "wild cards", [12] and [13] "early warning signals". Even if some of these concepts vary slightly, we consider all weak signals.

1.2 Challenges

What makes it difficult to detect weak signals is that they are generally hidden in the "noise of the daily produced data" [14]. This means that one of the main challenges that research in this area faces is to be able to distinguish something that can become an event from background noise that will eventually disappear in a large, continuously published data set. One of the traditional ways to deal with weak signal identification is to rely on human experts, who would analyze data sets and, based on experience, would make decisions on whether something qualifies as a weak signal or not. These services are often costly and not widely available. Even then, the increase in the volume of information may represent a challenge to the experts. On the other hand, early warning signals cannot be identified by using techniques on anomaly detection because they are usually not strong enough to be "accepted" as an anomaly or, even worse, other times they are part of the mainstream information that exists in the data set. So in that respect, techniques of anomaly detection cannot be directly applied; but some can be adapted and explored as research evolves in the area.

1.3 Contributions

There have been numerous attempts to detect weak signals even if this challenge remains recent. That is our motivation for summarizing these approaches. The contributions of this work are:

1. We raise a taxonomy of community detection methods based on various techniques used.
2. We present a qualitative panel of several definitions and techniques under different categories.
3. We compare the different detection techniques for weak signal detection.

1.4 Outline of the paper

This paper is divided into four main sections. Section 2 introduces the weak signal detection field, while Section 3 describes the theoretical framework behind weak signal representation of a future sign. Section 4 presents how different methods work over a standard data pipeline. Section 5 presents a taxonomy of the works in the field and discusses the state-of-the-art methods for weak signal detection and identification. Some conclusions are drawn in the final section in Section 6.

2 Early warnings landscape

2.1 The origins

[15] first introduces the concept of weak signal in the '70s to identify early strategic signals. He perceived that that, in general, companies does not react fast enough to early pieces of evidence of threats and opportunities over companies' strategic planning. Strategic planning is the process of "converting environmental information about strategic discontinuities into concrete action plans, programs and budgets".

According to Ansoff, a weak signal is considered as "graduated response through amplification". The proposed method contrasts with traditional strategic planning that relies on strong signals. For Ansoff, a weak signal has a high uncertainty, given that part of their point over inconsistent directions. In general weak signals can be detected with three filters [5]:

1. a surveillance filter to obtain the information
2. a mentality filter to indicate the relevance of the signal based on existing experience
3. a power filter to apply the learned knowledge in the decision-making process

One of the challenges of this review is to expose what is a weak signal. The intuition behind most definitions is that they represent unknown, unexpected or rare data over a specific period. However, the rarity makes it hard for this data to be identified as relevant, especially from a large volume of information that is continuously being received [16]. Weak signals may be early warnings of future changes and may forecast new possibilities. In fact, predictive research using weak signals is indeed an active field (17; 18; 19; 20; 21; 22; 23; and 24). We can advance briefly that weak signal detection implies the signal's nature (opportunity or threat) of the signal, its potential impact, probability, and delay before its occurrence (urgency). This can be classified as neither good nor bad and may threaten the current status quo or provide new opportunities for strategic activities. They are warnings that a new idea will change the present established environment, a novelty from the point of view of experts, and often opposite to their present opinion.

2.2 Definitions

2.2.1 Context monitoring and foresight

Context monitoring means scanning the environment and gathering information about events and their relationships (25 and 26). For [27], this also means providing foresight. Foresight is the capacity and ability to plan actions based on the acquired knowledge. For [28] the concept of weak signal detection only makes sense when it is inside a defined context. The context provides complementary information but narrows the attention focus and allows for a reduction of noise.

2.2.2 Data collection and organization

Several studies have been proposed to collect information, some focusing on an automated process by using a crawler (29; 30;31;32 and 33), others on knowledge of humans [34]. After collection, a clustering approach is commonly used to gather the number of identified signals and then identify weak signal clusters. Clustering may be semantic [35] or based on the knowledge structures of the data sources [32].

Adding a context to the collected data helps to improve its intensity, degree of precision, number, and liability [36]. [37] proposes a future signals sense-making framework (FSSF) based on environmental scanning and pattern management principles for collecting and organizing information.

2.2.3 Weak and strong signals

Ansoff categorizes signals into two groups: weak signals and strong signals. For [18], a weak signal is an alert warning "less visible and having a larger relative distance to the impact on future changes". Previous studies (38; 39) have also indicated that weak signals incompleteness does not accurately estimate of their impact. On the contrary, a strong signal remains easier to identify as its impact is clear, and stays consistent over time. A strong signal has the potential to become a trend, but a weak signal is a clue that may unveil how trends will evolve. The strong signal allows for a more accurate assessment of the situation, like tangible proof of a change [18].

It is essential to notice the fact that a weak signal may become a strong one. Indeed, the dynamics over different signals and their future impacts are non-negligible factors [20]. [16] characterizes a weak signal as "a factor of change hardly perceptible at present, but which will constitute a strong trend in the future".

The dynamics between weak signals that may become strong ones create a paradox [40]. While weak signal information increases over time, a weak signal can be regarded as a strong signal when the information permits accurate estimation of the impact. However, not all weak signals manage to become strong ones, and even if it happens, it may be too late for strategic decision-making [41]. The strategic paradox lies between waiting to have enough information

to perform better decisions or accepting the information's incompleteness and acting, accommodating the uncertainty that comes with it. According to [15], strategic issue management may help to solve this paradox. By continually monitoring the weak signals' evolution, decision-makers can plan a graduated and evolutive response on strategic issue management.

Here, we stated the weak and strong signals. It should be possible to correlate the management view of weak signals with the "weak signals" treatment in the signal processing field. In this sense, the signal-to-noise ratio of a signal could be more suitable to describe the strong and weak signals. A high signal-to-noise ratio indicates a strong signal while the opposite could point to a weak one.

2.3 Key features for understanding weak signals

2.3.1 Characteristics

[19], delves into the definition proposed by Ansoff with seven different characteristics :

1. an idea or trend that will affect how we do business, what business we do, and the environment in which we will work
2. new and surprising from the signal receiver's advantage point (although others may already perceive it)
3. difficulty in tracking down amid other noise and signals
4. various ways of interpretations of the same signal by the observer
5. improvement of its ability to thrive
6. often scoffed at by people who "know"
7. lag time before it will mature and become mainstream
8. representation of an opportunity to learn, grow and evolve

[21] features six aspects to describe a weak signal:

1. as phenomena of transition
2. the duration of a weak signal
3. its objectivity and subjectivity
4. various ways of interpretations of the same signal by the observer
5. strengthening of the signal
6. issues related to the receiver and analyst of the signal

These six characteristics help classify and understand the weak/strong signal dynamics.

2.3.2 Weak signals dynamics

[39] defines a deeper theoretical understanding of weak signals and proposes a 3D spatial model (i.e. signal, issue and interpretation) to define them, Fig. 1. The model is based on Pierce's model [7].

- *A signal* is the number of signals or their visibility. i.e., the number of times where the signal appears the signal means the visibility or the number of the future sign
- *The issue* is the number of events in which the signal appears. It indicates how far the future sign spreads.
- *Interpretation* represents the degree of recognition of future signals by information users, i.e., how many of those the receiver understand.

The magnitude of the future sign can be measured by a combination of these three dimensions [20]. A weak sign is a future sign which has a low level of signal, issue, and interpretation, meaning that a presentation of the future issue is about to diffuse to some extent but yet to be much interpreted by receivers. A weak sign becomes a strong sign when interpretation of the meaning of the sign becomes clear.

The combination of three dimensions allows to measure the the degree of a future sign. A weak sign is a future sign with has a low-level signal, issue, and interpretation left, lower, front part on Fig. 1. According to Hiltunen, these ones have rarely exposed topics. For this reason, they have a small representation, even if they can become strong. This definition is popular and extensively used by other researchers (34; 42; 43; 29;44; 45; 46 and 47). This semiotic model has become the standard to identify the dimensions of future signals.

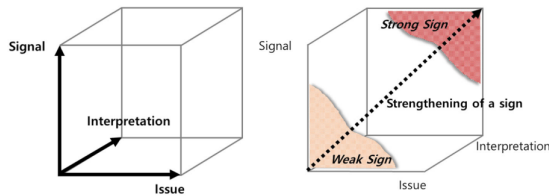


Fig. 1 Spatial model proposed by [39]

2.3.3 Weak signals correlation

A large number of existing studies have examined the notion of weak signals combination to reinforce their value and, in this way, identify specific events. [18] observed that despite the incompleteness and imprecision of the information, the combination of the weak signals could lead to the identification of an event. [20] identify weak signals as "current oddities (...) that are thought to be in a key position in anticipating future changes" can be combined with other signals. The combination of multiple weak signals provides more information for the identification of future events. A weak signal may gain expressiveness when combined with other signals. This signal becomes complete, which allows better planning by the firm ([5] and [35]). [48] characterizes weak signals as parts of a jigsaw puzzle building a holistic view of future changes.

2.3.4 Association with signal processing field

This paper elaborates weak signal detection from fundamental theory with Hiltunen and Ansoff to applications in different fields. It exists weak signal in Mechanical Systems, Signal processing and Big Data analysis. To the best of our knowledge, a survey on the subject combining weak signal in signal processing and weak signal in big data has not been reported yet. This intriguing research area attempts to extract weak signals of interest from noisy and complex signals in various fields such as wind energy [49], sonar [50], deep-space exploration and fault diagnosis of machinery [51]. In signal processing techniques, stochastic resonance (SR) is known to enhance weak unknown signals. SR can utilize the noise embedded in signals to extract weak fault characteristics from the signals ([51]; [52]; [53]). SR is an advanced signal detection method for early fault diagnosis of electromechanical equipment to ensure the reliable operation of electromechanical equipment and prevent malfunction in machinery fault detection [52].

2.3.5 Signal vs. Noise

One of the other challenges of detecting weak signals is distinguishing between signal and noise. The limit between noise and a weak signal can be confusing. For [54] the weak signal may be seen as "noise" where the value is not understood before the appropriate treatment. This kind of signal is difficult to monitor and can be confusing. Mendonça points out that "weak signals can be conceived as outliers (anomalies) that do not easily fit with the understanding embedded in those coherent futures constructs of foresight project" [18].

Searching for relevant data for event anticipation presents different challenges: detection complexity, identification and interpretation. Generally, the procedure for forecasting or early warning scanning consists of four steps [55]:

1. exploring the weak signals
2. assessing the weak signals
3. transforming the signals into issues
4. interpreting the issues for new futures

In this regard, this paper presents the different methods to detect and identify weak signals in textual data sets.

3 Early warning signals' theoretical framework

Analysts may tend to ignore information that does not conform to their own biases or beliefs and, at the same time, underweight risks resulting over the information they consider certain [56]. As a result of analyst bias, companies may miss, or refuse to see, changes in the market as they cannot realize a future based on different paradigms. This self hurting denial is not a new phenomenon; for example, at the beginning of the last century, ice traders already failed to forecast the ice trading market decline [57]. A more recent example is Blockbuster that had the opportunity to buy Netflix, and refused

it, as it did not anticipate the changes in the video market [58]. Furthermore, Kodak invented the first digital camera in 1975 but ultimately lost the digital cameras' revolution, leading it to file for bankruptcy in 2012 [59]. Kodak's case is intriguing, as it shows that even though analysts may have the information, they may choose to base decisions over the parts of information that support their own, possibly adverse to changes, opinions.

One of the early motivations for the study of weak signals is developing strategic planning for companies. Having tools that help forecast, in an unbiased way, changes in the market is crucial. For this reason, the weak signal weak is popular in business environments [22]. The examples of applications go from predicting future energy trends [47], for monitoring the use of health resources applied, for treating chronic diseases [60], passing through forecast the evolution of nanotechnology [43]. However, it is also used to follow and forecast the evolution of other fields such as school bullying [46] and tourists volume and flow [61], for example.

3.1 Weak signals dimensions

Different dimensions may be used to measure the importance of a signal. For example, we can measure a signal regarding its frequency, temporality, impact, rarity and unrelatedness paradigm. In this section, we will see some examples of how different researchers have quantified signal dimensions.

3.1.1 Visibility

In general, weak signals present low visibility [8]; they are the kind of signal that may discreetly pass without being remarked even by specialists. If we consider the research done over the analysis of texts, visibility could be measured by the terms' frequency. Term frequency is the rate of occurrence of a word in a series of texts and can be measured by counting the number of occurrences of a term in a document [62]. [63] use betweenness centrality for convergence measurement and minimum spanning tree (MST) as a way to put in evidence the essential relations.

Based on Hiltunen's concept, [29] proposes new metrics to quantify the signal axis. The degree of visibility (*DoV*) and the degree of diffusion (*DoD*). To evaluate the degree of visibility of a signal, Yoon proposes measuring the frequencies of keywords. *DoV* measures the degree of a defined keyword's frequency in a set of documents (section 3.1.1). The author proposes two measures for evaluating emergency and the issue impact of a signal. These two indexed maps are called keyword emergence map (KEM) and keyword issue map (KIM). Yoon focus on two main principles:

Proposition 1 *The number of occurrences of a keyword is linked to its influence*

Proposition 2 *Recent occurrences of a keyword present greater interest than older ones*

For [29], the reasoning behind the weak signal detection lies in searching for keywords with a low occurrence frequency and a high, increasing rate. These two factors are markers that can be used to detect weak signals. The first part is related to the visibility of the signal, and the second with its diffusion. Keywords exhibiting a high occurrence frequency and high, increasing rate are markers of strong signals. While both, weak and strong signals, expose growing visibility, the total number of occurrences for weak signals is still lower when compared to strong ones.

3.1.2 Diffusion

Diffusion is how Yoon classifies Hiltunen's issue axis, and it represents the spreading of the signal. Yoon's metric to detect an emerging issue is the measure of keyword frequencies, the degree of diffusion (*DoD*) [29] of a keyword. *DoD* is complementary to the *DoV* metric. *DoD* focuses on the issue-level of a future sign, especially with the increased document frequencies of keywords. Yoon measures each keyword's frequency regarding the total number of documents in the portfolio maps method [29]. Note that [8] also highlights that the signal strength increases with the number of issues.

3.1.3 Interpretation

The theoretical foundations of the weak signal definition proposed by [39] convey an interpretation dimension. For Hiltunen, this dimension includes the context aspect of how people forecast possible future events. At first glance, a weak signal is an incomplete and fragmented piece of information, so uninterpretable as it is. Strong signals, on the other hand, are more precise and superior in quality. Contrary to Hiltunen, [45] tried to address this feature by employing keyword clustering and topic selection based on keyword co-occurrences.

3.1.4 Rarity & paradigm unrelatedness

The main characteristics of a weak signal, as proposed by Hiltunen, constitute a weak signal representation. [64] emphasize the fresh nature of signals with an approach to novelty-focused weak signal detection.

3.2 Challenges

3.2.1 Weak signals detection

Whiteout a doubt, the main challenge for all the methods discussed in this article is identifying a piece of information as a weak signal. A fundamental characteristic of a weak signal is its rarity and incompleteness, which makes it hard to detect as it can be confused with noise. However, the difficulty comes from the complexity in characterizing small and spread pieces of information as relevant. The problem also resides in recognizing their forecasting properties as they announce future events. Mine, the relevant information buried under

excessive and contradictory noise, consistently and reliably, is the very essence of the weak signals detection field.

3.2.2 Database consistency

According to [65], [66] and [67], newspapers, the internet and especially blogs are good sources to find weak signals. Moreover, the correlation of some of these sources can help put in evidence tendencies that are not clear when just looking into one single source of information. [64] used two databases: a referral database and a study database. The choice of the databases is crucial as they significantly affect the results of the analysis. These databases have to be coherent and meaningful regarding each other. Moreover, they need to be complementary and provide additional information. If these characteristics are not respected, hardly one could highlight any relevant information, even when correlating different data sources.

3.2.3 Impact

For Ansoff, a weak signal is "warnings(...), events and developments which are still too incomplete to permit an accurate estimation of their impact". Impact estimation consists of identifying the potential consequences of a signal and estimating its importance. [34] provide a method for cross-impact reasoning to infer future events' likelihood and measure the strength of influence. This method is based on an inference model using the Bayesian network where a node represents an event, and a link between two nodes represents the impact probability of a state transition from one node to the other.

3.2.4 Interpretation

In general, at first glance, it is difficult to interpret a weak signal. However, the interpretation may become more straightforward and explicit when a weak signal is combined with context and/or other weak signals. According to Hiltunen, the "interpretation" dimension is the assumption or sense of people regarding future events (subjective matter). The authors in [68] emphasized the limitations of Yoon [29]'s approach (presented in part 5.1.1) based on the disappearance of some signals and the complexity of interpretation. This shows the difficulty to measure this dimension.

3.2.5 Weak signals duration

[21] raises some questions regarding weak signal duration. For him, studies that consider this dimension of the problem can be divided into two categories: (1) the weak signal only lasts for a moment, or (2) the weak signal lasts longer. The underlying assumption is that the weak signal is either the phenomenon (Assumption 1) or a sign of a change (Assumption 2).

4 Weak signal detection pipeline

This section contextualizes contextualizes weak signal detection over a standard data treatment pipeline, a simplified version of CRISP-DM (CRoss Industry Standard Process for Data Mining) process [69]. The data treatment phases taken into account are data collection, preparation, modeling and evaluation, as shown in Fig. 2

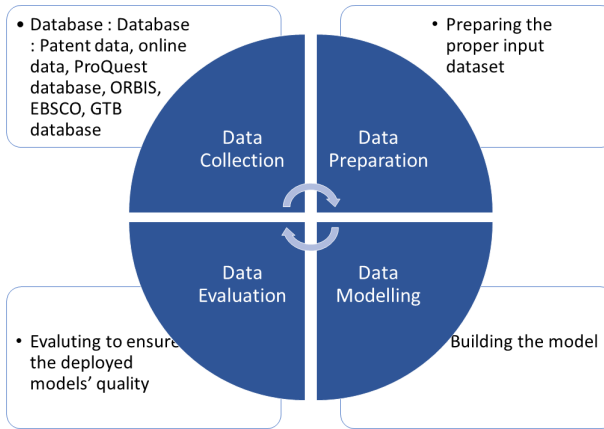


Fig. 2 Data Process for data treatment pipeline produced

4.1 Data collection

Data acquisition is the process of gathering data and measuring information on variables of interest. This step is the most basic one; without data, no signal can be processed or detected. The quality of the data collection phase directly impacts on the overall result of the proposed system (70 ; 71). It entails the definition of the kind of data is needed, the evaluation of the data sources reliability, the acquisition of the data and the quality verification of the received information.

Authors use different methods to perform data collection. Traditional data sources are newspapers, scientific journals, patents, specialized data sets, social networks and the Internet in general, in special blogs. Some authors propose the of web crawler and search queries (29; 47; 46; 44; 30; 64; 72). [34] propose Global Trends Briefing (GTB)¹. GTB is a network of experts, from different domains and spread internationally, proposing documents of futuristic interest. Documents were collected, analyzed and validated before the integration into the futuristic dataset.

¹Korea Institute of Science and Technology Information, <http://www.kisti.re.kr/english/>.

4.2 Data preparation

Data preparation is the technique used to transform the acquired data into a useful dataset [73]. It covers an extensive range of treatments applied to the raw input to fix errors and create a valuable dataset for the next phase. Raw data is often imperfect, inconsistent and redundant. Data preparation, or preprocessing, methods solve all the main problems linked to the data itself and adapt to the modeling phase's requirements, improving its effectiveness.

Examples of treatments that can be done are dealing with missing or incomplete information [74], noise [75], and removal of duplicates and data formatting. The operations vary according to the type of the raw acquired data and the main objectives of the whole pipeline. A regular process in the data processing phase is removing stopwords and performing general data formatting. In the processing phase, several authors use feature selection [76], feature scaling and engineering to identify the data characteristics such as "name", "age" or "gender" [77]. The missing values can be treated with different mathematical tools like normalization, min-max technique, average, median value, and other missing value-replacing methods.

4.3 Data Modeling

Modeling consists of choosing the most fitted models and their parameters to reach the expected result. The number of possibilities is significant, and it is in this phase that the primary intelligence resides. It is where the signals are understood as real signals, and their relationship is revealed. There is a close relation between modeling and data preparation. Often the data understanding process consists of finding problems in the modeling phase, returning to the preparation to create, or format, new data to be used in the modeling phase [69].

As an example of techniques used on the model, phase are: processed for interpretation and pattern recognition (linear discriminate analysis, support vector machine, neural networks, decision tree, k-nearest neighboring method, for instance).

4.4 Evaluation

Having the results of the modeling phase is not enough. The found weak signals need to be evaluated to ensure the deployed models' quality and efficiency. This phase tries to verify that the reached results fit the task's description, if parts of the answers were not considered and if the process meets the quality standards required regarding false positives and false negatives.

5 A categorization of weak signal detection techniques

This section presents some methods proposed for weak signal detection and tries to organize the strategies into different categories as shown in Fig 3. The

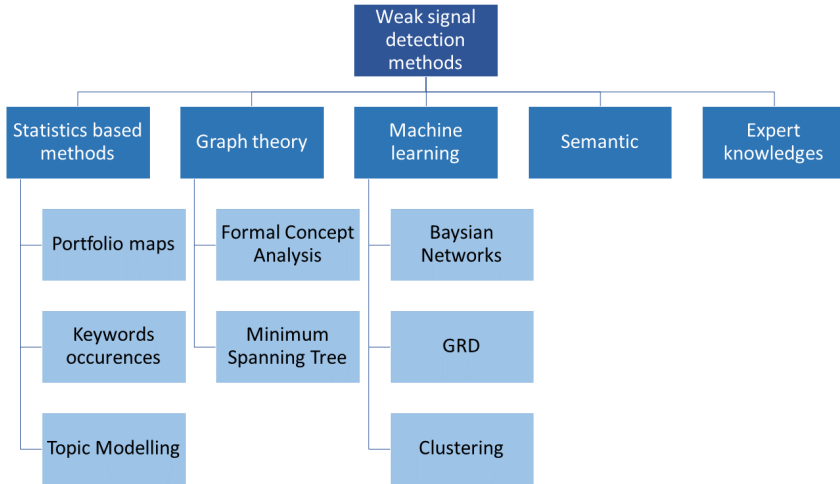


Fig. 3 A taxonomy of different methods of weak signal detection

objective is to highlight the different techniques used on some representative techniques. For example, several authors use portfolio maps methodology with two portfolio maps to extract the potential weak signals. Furthermore, [63] and [31] use methods based on graph theory. [64] propose to observe the rarity and the paradigm unrelatedness of a weak signal. [64] and [42] consider the interpretation dimension of a future sign. [43] have a more text mining-oriented approach, based on the word frequencies. The discussed methods contribute to providing information on weak signal detection. The way the methods characterize and handle a signal varies.

Table 1 and Table 2 summarise the main of the evaluated methods. They expose the essence of the studied methods providing an easy way to compare the methods and their strategies. The columns of Table 1 and Table 2 and consist of:

- **Aim:** The objective of the work, i.e, the problem it is trying to solve
- **Data used:** Kind of data used for the analysis
- **Applications:** Field of application of the work
- **Scoring metrics:** Field of application of the work
- **Keywords selection:** Uses keyword search to find documents
- **Anomaly detection:** Considering weak signal as an outlier
- **Clustering:** Uses Machine learning techniques for group unlabeled examples
- **Information granularity:** The granularity of the treated data, whole document or keywords
- **Temporality dimension:** Considers the temporality dimension during the weak signal detection
- **Finding relations and patterns:** Tries to discover links between elements and determine a design that is repeated at regular intervals

Table 1 Comparison of some classic community weak signals detection

Papers	Features	Aims	Data used	LEA	Applicat	Metrics
[30]		Prediction of future organised criminal threats	web sources, social media and Reports	C	TF	
[77]		Modeling crime problems	Crime data from a sheriff's office	C	×	×
[31]		Prediction of future organised criminal threats	web, social media	C	×	×
[78]		Prediction of future lone wolf	web sites and forums	C	×	×
[63]		Detection industry service trends	existing database	E	co-occurrence matrix between phrases (K)	×
[79]		Detecting changes for businesses	Not given	E	×	×
[33]		Investigation for renewable energy	storage technologies for renewable energy	En	×	×
[47]		Forecasting nuclear energy events	news	En	DoV & DoD	
[80]		Forecasting	Academic papers	IT	×	×
[29]		Detecting business opportunities	Web news articles	IT	DoV & DoD	
[72]		Smart grids problems	Academic papers	IT	DoV & DoD	
[45]		Foreseeing the advent of new technologies	Academic papers	IT	DoV & DoD	
[42]		Forecasting changes for remote sensing	newspaper, paper, social media	IT	DoV + DoD	
[64]		Technology foresight	social media content, news	IT	rarity & paradigm unre-latedness	
[43]		Predicting the innovation	Academic papers	IT	annual growth rate and standard deviation	
[46]		School bullying issue forecasting	social media	S	DoV & DoD	
[44]		Land administration sector issue detec-tion	annual	Admn	DoV & DoD	
[81]		Forecasting	Academic papers	Med	intra indicator and inter indicator	
[34]		Foresighting future technologies	academic papers	GF	Knowledge Matrix	

Papers	Features	K-Selection															
		Anomaly Detection	Classifier	Clustering	Information granularity	Temporal dimension	Pattern recognition	Interpretation dimension	Semantic consideration	Text-mining based approach	Topic Assignment	Data Fusion	Statistical model	Graph theory	Bayesian Networks	Multi-words analysis	Primary metrics
[30]		Y	N	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	N	TF	N
[77]		Y	Y	K	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N	N
[31]		Y	N	K	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N	N
[78]		Y	N	K	Y	Y	Y	Y	Y	Y	N	N	Y	Y	Y	TF	Y
[63]		N	N	K	N	Y	Y	Y	Y	Y	NA	N	N	N	N	TF	N
[79]		N	N	N	N	Y	Y	Y	Y	Y	NA	N	N	N	N	TF	N
[33]		Y	N	Y	Y	Y	Y	Y	Y	Y	N	N	N	Y	Y	TF + DF	N
[47]		N	N	K + D	Y	Y	Y	Y	Y	Y	C.A	N	N	N	N	TF + DF	Y
[80]	NA	Y	N	K + D	Y	Y	Y	Y	Y	Y	N	N	N	NA	N	N	Y
[29]	Y	N	N	K + D	Y	N	N	N	Y	Y	C.A	N	N	N	N	DF	Y
[72]	Y	N	N	K + D	Y	Y	Y	Y	Y	Y	C.A	N	N	N	N	DF + DF	Y
[45]	Y	N	N	K + D	Y	N	N	Y	Y	Y	TI	N	N	Y	Y	DF + DF	Y
[42]	Y	N	N	K + D	Y	Y	Y	Y	Y	Y	N	N	N	Y	Y	DF + DF	Y
[64]	Y	Y	N	K + D	Y	Y	Y	Y	Y	Y	N	N	N	Y	Y	DF + DF	Y
[43]	Y	Y	N	K + D	N	N	N	N	Y	Y	N	N	N	N	N	TF + DF	Y
[46]	Y	N	N	K + D	Y	Y	Y	Y	Y	Y	N	N	N	N	N	TF	N
[44]	Y	N	N	K + D	N	N	N	N	Y	Y	C.A	N	N	N	TF + DF	Y	Y
[81]	Y	N	N	K	N	Y	Y	Y	Y	Y	N	N	N	Y	Y	TF + DF	Y
[34]	N	N	Y	K	N	Y	Y	Y	Y	Y	N	N	N	N	N	TF	Y

- **Dimension Interpretation:** The way the method express the receiver's meaning
- **Ranking data:** Way to sort data based on frequency criteria
- **Semantic Consideration:** Considers the meaning of knowledge using semantic contexts and background knowledge
- **Text-mining techniques:** Uses natural language methods for exploiting unstructured documents
- **Topic Modeling:** Uses automatic topic modeling to classify documents
- **Topics Assignment:** Assigns a topic name to a set of words
- **Data fusion:** Integrates multiple data sources
- **Statistical method:** Uses models and techniques based on the statistical analysis of raw research data to extract information
- **Graph Theory:** Makes use of graph theory-based approaches
- **Bayesian Networks:** Uses probabilistic graphical model using Bayesian inference
- **Multi words Analysis:** Considers group of words instead of single keywords
- **Primary metrics :** Which is the basic metric used on the work, e.g. term frequency and document frequency
- **Scoring metrics :** Which is the indicator used to quantify the interest of signal
- **Maps :** Makes use of portfolio maps technique to classify the keywords

5.1 Statistics-based method

This section presents propositions that use statistical approaches to detect weak signals. The methods are further group the works on this class into sub-classes, presented in Fig. 4.

5.1.1 Portfolio maps method

The methods in this group are categorized by constructing portfolio maps and identifying topics during the weak signals analysis as shown in Fig. 5. Most initial methods base their weak signal detection method over statistics linked to the keywords on initial data. One of the first proposals is the portfolio maps' methodology proposed by [29], followed by other authors such as [68], [72], [46], [45], [44],[47],[43], and [42]. These works, in general, consider Hiltunen's signal classification (section 2.3.2) as a way to measure the importance of the detected signals.

Initial works

In the original work, [29] suggests a quantitative approach based on keywords text-mining to identify weak signal topics in the business opportunities on the solar cells. The work objective is to minimize the intervention of experts on the detection and identification of weak signal topics. For the data collection, Yoon relies on news websites and the ProQuest database (<http://search.proquest>.

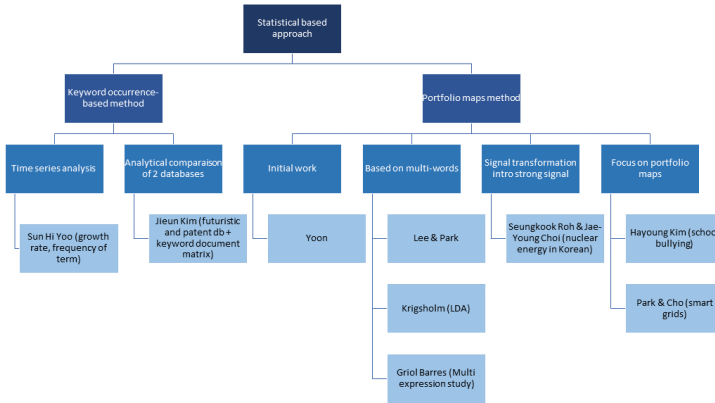


Fig. 4 Taxonomy for works using statistical methods produced

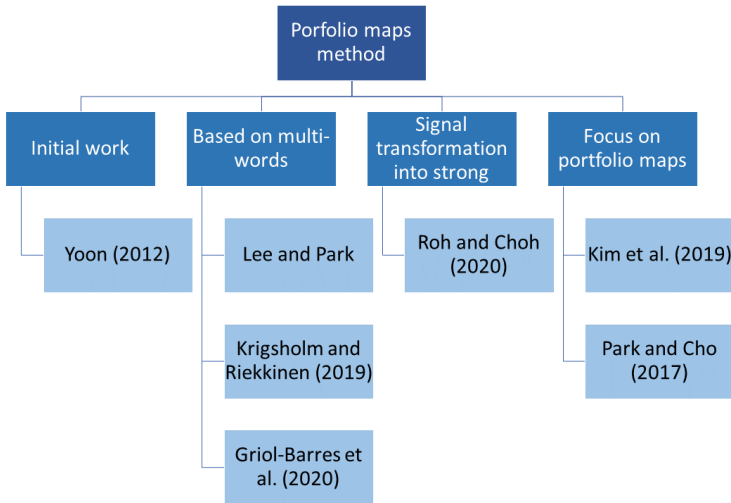


Fig. 5 Taxonomy for works using portfolio maps methods

com), including various information sources on different subjects (political, economic, social and business, technological). A total of 28270 English Web news articles were located using the keywords "solar cells" and "photovoltaic", for the period from Jan 1, 2006 to Mar 31, 2011.

For Yoon, weak signals are imprecise indicators, rarely considered by people, but that announce future events. The developed method proposes two new metrics, Degree of Visibility (DoV), and Degree of Diffusion (DoD), as indicators to analyze the detected signals. DoV , equation (2), corresponds to the frequency of a defined keyword in a set of documents (signal axis), DoD , in

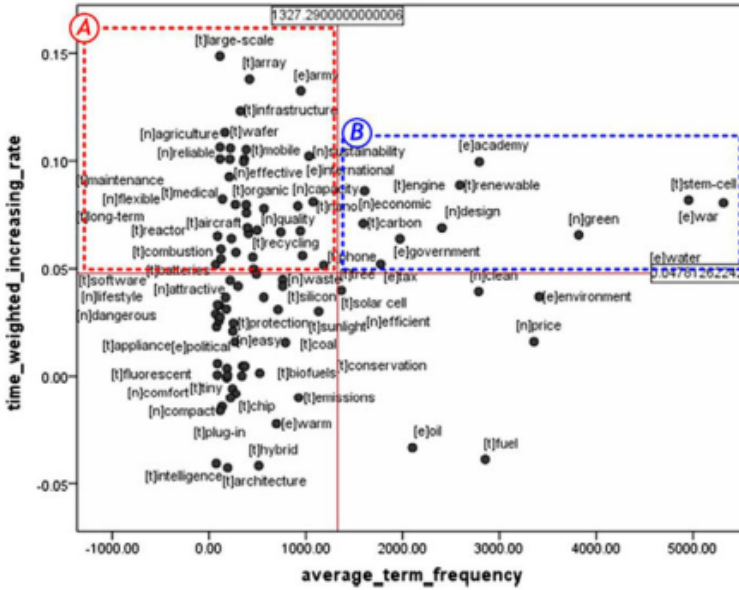


Fig. 6 Keyword emergence map - In area A, in area A are the keywords that are connected with the weak signals as presented in [29]. In the portion B, keywords are connected with the strong signals.

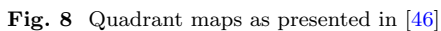
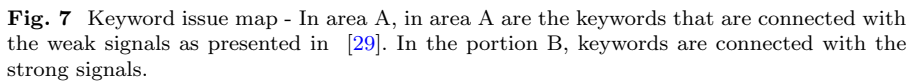
equation (1), represents the document frequency of each keyword.

$$DoD_{ij} = \left(\frac{DF_{ij}}{NN_j} \right) \times \{1 - tw \times (n - j)\} \quad (1)$$

$$DoV_{ij} = \left(\frac{TF_{ij}}{NN_j} \right) \times \{1 - tw \times (n - j)\} \quad (2)$$

where TF_{ij} is the frequency of keyword i in the period j , DF_{ij} is the document frequency of keyword i in the period j , NN_j is the number of considered articles in the period j , n is the number of periods, and tw is a time-weight which is linked to the changing speed of the target topic.

DoD and DoV metrics are designed in two keyword portfolio maps (KEM - Keywords Emergence Map, Fig. 6, and KIM - Keywords Issue Map Fig. 7 and Fig. 8) to gauge the importance of the keywords extracted from the text mining. Each map is composed of four quadrants that group keywords into weak signal, strong signal, latent signal and well-known signal first mentioned in [68] followed by [72], [45], [44] and [46]. The vertical dimension refers to the average increasing rate of DoV (or average growth rate for DoD) and the horizontal dimension the average term (or document) frequency. The step of identification of weak signal topics is based on the previous representation of a weak signal. Weak signal words have a lower current frequency but higher growth rate, which suggests that they can increase quickly. The keywords related to weak



signal topics are situated in area A of KIM in Fig. 7 and KEM in Fig. 6. One of the advantages of this method is to automatically and dynamically categorize the keywords. Based on the maps, experts can then interpret the results (82 and 83). According to Yoon, the proposed method results better than human experts when dealing with massive textual information.

The authors in [68], [72], [45] and [44] raise the uncertainty problem and lack of interpretation that occurs when keywords appear in different quadrants of KIM and KEM.

Keyword portfolio map is a popular method and used by authors in different ways. For instance, [45], [44] and [42], introduce the use of multi-words analysis. Their hypothesis is that working on a single keyword can penalize the understanding of the specific subject.[47] have searched to verify the validity of portfolio maps' method analyzing the evaluation of signals from weak to strong.

Study based on multi-words

[45] explain that the use of a single keyword may lead to a loss of objectivity. Two factors can explain that, the first is the lack of distinction when the signals are mixed at each map. Indeed, in the original proposal in [29], the objective was to allow experts to decide whether the keyword should be discarded or not. The other limitation concerns the multiple numbers of meanings for a given keyword. According to [68], [72], [45] and [44], interpretability is lost when performing the analysis. Moreover, the keywords related to weak signals are generally isolated terms. When performing a deeper analysis the lack of relations and context over the keywords limit the information. One of the suggestions of these works to improve interpretability is to include semantic analysis in future signal detection frameworks. To decrease the uncertainty problem, [45] propose working with a group of words and concepts instead of the isolated keywords, which should increase the whole system's accuracy.

The analysis of [45] and [44] present some similarities. For example, the weak signal detection pipeline collects and prepares data, builds map and identifies topics. The collected data from EBSCO host database ¹ and Scopus, respectively. Using queries correlated to their fields, ethnical issue in AI for Lee and Park and land administration for [44]. They select relevant keywords by applying a criteria-based method on the apparition frequency of a word. [45] employed a handmade list as to the main criteria for keyword selection, while [44] used a combination of automated and manual selection. [45] assess the degree of interpretation by employing keyword clustering. They build the maps for emergence and issue, focusing on a set of commonly located keywords in the same quadrant in the KIM and KEM. They inspect each cluster's keyword and confirm whether the group should be a topic.[44] use the Latent Dirichlet Allocation method for identifying the topics. The majority of the keywords are classified as either strong or latent signals. The reason is that the average values are used as thresholds. Intuitively, the number of topics considered is a trade-off between the simplicity of interpretation and coherence regarding the knowledge domain. They pay attention to the top fifteen most common terms from each topic.

Then, They verify the keyword status in two portfolio maps. After this step, a name and a group of keywords were associated with the topics. However,

¹<https://www.ebsco.com/fr-fr>

some topics have common words, thus sometimes merged. In the end, nineteen topics were considered. The topics identification under four signal categories was based on the following rule: the category was determined by in which quadrant the most important keywords of the topic landed. [45] find three weak signal topics on eighteen topics and [44] obtain one weak signal topic on nineteen.

These approaches do not consider the evolution of weak signals into strong ones. A more systematic analysis over different and successive periods of time could reveal topics' evolution. Another potential approach could be discovering semantic networks in patterns clustering keywords and study their evolution (33 and 84).

[42] focus on the analysis of heterogeneous and unstructured information. The authors consider weak signals as small and seemingly insignificant issues, but that can provide changes in the future. They vary the number and types of sources (ScienceDirect, New York Times and Twitter) and automatically extract the documents' keywords. The authors use the Degree of transmission (*DoT*) metric to help in the evaluation of the importance of the terms. This degree considers all h-index values from the journals from all texts where the word i can be found, as shown in Equation (3).

$$DoT_i = \sum \text{Hindex}_{\text{journal}} \quad (3)$$

For the final consideration of terms related to weak signals, every *DoD* and *DoV* are multiplied by their *DoT*. They observe that some word expressions are identified as potential weak signals. In order to provide more accurate information, authors apply multi-word analysis with Natural Language Processing. In this way, they investigate the percentage of times the word appears beside a potentially weak signal term in all the dataset documents. They exemplify this analysis by the term "desertification", and show other terms that relates to it, for instance, "processes" with 36.76%, "Areas" with 26.37% and Rates with 9.58%. A possible evolution for this method would be consider the word embedding [85], which can help to identify related and similar terms over the analyzed documents.

The authors identify two assessments for the evaluation: identifying strong signals and a group of experts. With these two assessments, the methodology shows interesting and promising results. More generally, this work shows, on the one hand, the validity of the portfolio maps method and, on the other hand, an improvement in the integration of the dimension of interpretation. Another additional point is that the system is independent of its input dataset of documents. In other words, the research work can be applied to any field. However, it is crucial to notice that the H-index does not necessarily give an accurate ranking of the impact. It merely states how many times a scientist has published and how many times it has been cited. The h-index puts newcomers at a disadvantage since both publication output and observed citation rates will be relatively low. A young scientist will score a low H-index, because they

do not have time on their side. H-index is based on rather long-term observations. Another drawback of using h-index is the temptation for researchers and journals to boost their H-index through self and arranged cross citations.

Other works have rather focused on the topic modeling and not on the categorization of signals. For instance, [86] work from topics and [81] with topic modeling Latent Dirichlet Allocation (LDA) in section 5.1.4.

Signal transformation study

This section presents methods to evaluate the possible evolution of weak signals into strong ones. Not all weak signals manage to become strong, but detecting early the ones that may become is an advantageous for strategic planning, as it may conduct foresight [27]. However, the dynamics may also create a strategic paradox. Either one waits to obtain sufficient information to act better or accepts the information's incomplete nature to act quickly. Weak signals are early indicators of change, but not sufficiently reliable for strategic decisions. Acting only when they become strong signals, may imply that it is too late for them to represent a strategical advantage. At this point, the signal has already become general knowledge.

[47] propose a method for detecting the evolution of a sign. They wanted to prove the validity of the portfolio map's method to detect the transformation for future signs. [42], in their turn, also consider the transformation of the weak signal into strong in the evaluation step. These studies propose two different approaches to show the transformation of the weak signal. [42] highlight the process in the evaluation stage and [47] develop a robust analysis. [42] identify two assessments. The first is based on detecting strong signals in a new input dataset with more recent documents. The second assessment is based on a group of experts. In the first evaluation process, the strong signals in recent documents that were detected as weak signals in earlier references confirm the process.

[47] target the prediction of Korea's nuclear situation post-2019. The input dataset consists of news articles with the term "nuclear energy" from the site "Naver News" <https://news.naver.com/> from 2005 to 2018. Articles are grouped into six months periods. The frequency of occurrence determines the pertinence of a word. A word is considered as a keyword if it appears at least once at each six-months interval. On the evaluation of the results, two hypotheses in [47] are raised to verify the Yoon's methodology :

1. Can the weak-signal methodology predict the UAE nuclear contract by using social media data from the first half of 2005 to the first half of 2010?
2. Is it possible to predict the Korean government's nuclear phase-out declaration using social media data from the first half of 2011 to the first half of 2017?

They apply the portfolio maps method over two different time intervals for each hypothesis. The first one allows the detection of weak signals, while the second is used to evaluate the evolution of the signal from weak to strong. For

the first hypothesis, the chosen intervals are from 2005 to 2009 and 2005 to 2010. The presented results point to the direction that using portfolio maps is a reliable method, even if no definition over the expected time for the transformation of weak to strong signal is provided. For the authors, a determining factor is an increase in appearance rate, even more than the term frequency. Their work argues that an increased rate of 0.3% indicates that the signal may become a strong one in the future, even if not verified with experimental data.

Other applications

Some studies apply portfolio maps to gain understanding and follow the evolution of a series of distinct topics. For example, [46] target the identification of school bullying in South Korea through weak signals detection. [72] look for smart grid issues using the future sign searching technique.

Traditional social-based methods are not enough for detecting new trends of school bullying. For this reason, [46] rely on big data analytics to understand the evolution of school bullying. Their method relies on online data (blogs, Twitter, social networking websites and 257 news sites) in which the term "school bullying" appears. Based on portfolio maps (KIM & KEM), some topics show a high increasing rate compared to other words. These have a high likelihood of becoming more socially impactful words. The work results conclude that bullying, group assault incidents and physical violence cases are perceived as extremely serious among youth.

[72] look for smart grid issues using the future sign searching technique through text mining. For validating the method proposed, the authors collect documents at different points in time by using search queries ('smart grids') over Radian6 (social media monitoring platform for marketers). In this investigation, the authors do not consider the transition of a weak signal to a strong one. Indeed, working on this problem means finding a suitable time for a weak signal to turn into a strong signal. It is decisive to notice that the conversion of time (weak signal into a strong signal) is uncertain and may vary depending on the study's subject. The work uses the portfolio maps' approach to track changes during the transitional period. It would be possible to find the converting pattern from a specific signal into another signal in a time series (converting a weak signal into a strong signal) and how general keywords go through the quadrants. A further study would be to change the window time. Here, the entire period is the year, but if the future work changes the time window's size, it would be possible to monitor a group of keywords.

5.1.2 Keywords-occurrence based method

The main fundamental characteristics of (64; 29; 46; 44; 45; 47; 42 and 43) are the use term of frequency. The following studies (43 and 42) present additional contributions to the detection of weak signal based on frequency.

[43] propose a study to predict innovation and explore the dynamic behavior of weak signals using the agent-based simulating tool NetLogo. Thus, the authors present a time series analysis of words based on the word frequencies

through text mining. This study's technical challenge involves identifying the concepts that have a strong possibility of being weak signals. Authors consider weak signals as emerging topics related to keywords that were not selected. In other words, a word that has an unusual growth rate could relate strongly to future unfamiliar and unusual issues. They collect 50 articles from Journal of the Korean Ceramic Society with search queries between 2008 and 2012. The proposed method computes each word's occurrence, an annual occurrence, and an annual growth rate. Here, they use the frequency of a term as visibility of a signal ("signal" axis from Hiltunen's definition) and the growth rate of a term as a description of diffusion ("issue" axis). Their concept of a weak signal is consistent with Hiltunen's definition. The hypothesis is that a weak signal appears rarely and can become stronger when merged. Their research focuses on infometric analysis; this allows the reinforcement of the expert-based detection method. Then, by applying these terms to each agent of the agent-based model's values, authors could extract the future strong signals. The model represents another method of identifying weak signals, but it would be interesting to determine an optimal threshold.

5.1.3 Analytical comparison of two databases

Another research that has provided evidence for weak signal detection is 64. They explore a direction to determine the degree of novelty of documents in a dataset. In this way, they consider a weak signal as a novel future signal (documents or keywords). The target research field for this work is futuristic analysis on the Augmented Reality. Their technique uses different futuristic data types to determine which documents could represent a significant future trend. As proof of concept, they used two datasets, an online dataset used as of weak signal's source and a patent dataset used to access existing paradigms of technological innovation. The online futuristic dataset is a collection of articles from different technology foresight websites, the dataset that works as a technology proxy is a collection of patents linked to the Augmented Reality field. The method consists of standards steps, the first is the data collection. As documents are unstructured, the second phase is to extract the keywords of documents and them into keyword-document matrixes, one for each dataset. The third phase is the classification of the documents related to their rarity. The fourth step is to determine a signal-portfolio map to access document patterns and find weak signals. Documents on both datasets are ranked on behalf of two parameters: rarity and paradigm unrelatedness. Authors define rarity as the rate over which technology foresighters do not commonly mention the signal. It represents how unusual or infrequently seen the idea is, compared to others in the same set, i.e., the infrequency of an idea [87]. Paradigm unrelatedness is considered as the rate at which a signal is unrelated to previous technology innovations.

Document-level and keyword-level analyses are implemented as the technique considers as a signal for both documents and keywords. The method proposes using the Local Outlier Factor (LOF) [88] to organize the articles

and find out similar clusters. One of the advantages of LOF is that it filters out local outliers, as it analyses the document's density compared to its neighborhood. LOF is based on the k -nearest neighbors. In this context, an outlier is an individual having a lower density than their neighbors. The intuition is that a new item will be seen as an outlier. LOF is calculated for both levels, documents and keywords. Then, the rarity assessment and paradigm unrelatedness are achieved with a signal-portfolio map. The map is built using rarity and paradigm unrelatedness in Fig. 9 as two axes and locate signals, i.e., documents or keywords.

This model presents a new approach for the detection of weak signals. It uses two distinct datasets to cross-check the importance and impact of the themes on both. One of the main advantages of this work is that it can be completely automatic. All the steps can be done without human intervention from the data collection to the signal-portfolio map. The keyword extraction takes charge of the tasks of defining the available themes. The most important part of the method is LOF's application to assess the rarity and paradigm unrelatedness of the themes. On the one hand, the definition proposed by Kim and Lee provides additional information to the standard definition, and on the other hand, the LOF detection method shows the outliers, which can be weak signals.

This work of research relies on the complementarity of the two chosen datasets. One is used to access the technology (online data: posts from weblogs, news, databases, discussion forums, Wikis,...) and another to evaluate the futuristic potential of these technologies (patent db - UPSTO). These datasets must be carefully chosen, as the whole method relies on their characteristics. The first step of the LOF consists of determining the K number of neighbors. The definition of the value of K is not straightforward and directly influences the final result. In fact, K 's best value may vary regarding the size of the collection or the considered documents' quality. The method does not create a forecasting model *per se*, it relies on the entire input datasets to produce a result. It needs to keep all the analyzed data in memory what may be a problem for large datasets.

5.1.4 Topic Modelling

[81] and [44] have investigated the use of topic modeling to detect weak signals as shown in Fig. 10. [44] questions some past hypotheses about Yoon's method. That is why they combine Latent Dirichlet Allocation influences with Yoon's categorization.

[81] propose a methodology for weak signals detection using LDA and o verifying the study feasibility. For the authors, a weak signal is characterized by a small number of words per document, present in a few documents, meaning rarity and abnormality. Words that present a close connection in a given theme are unrelated to other themes but appear in similar contexts. Note that the authors do not consider the temporal aspect of documents. Their approach is based on clustering topics at multi-level documents and extracting significant

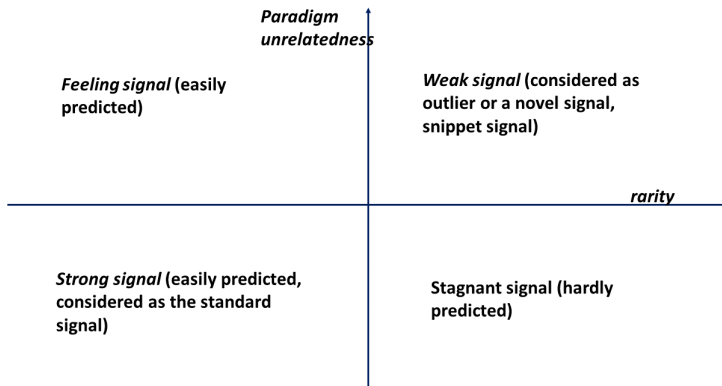


Fig. 9 Signal-portfolio map based on the rarity and paradigm unrelatedness proposed by [64]

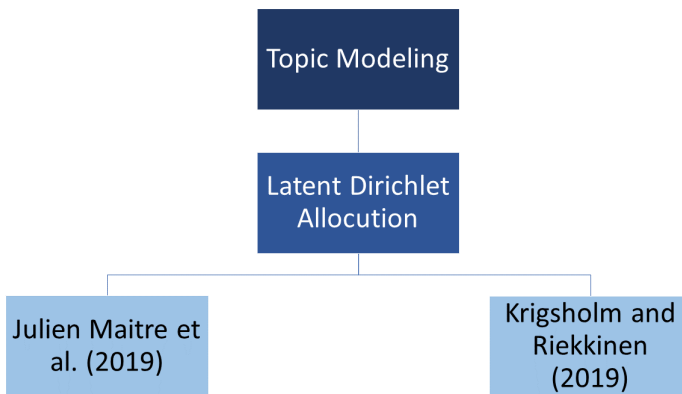


Fig. 10 Taxonomy works on topic modelling's method

descriptors (weighted list of words). In the study, [81] combine Latent Dirichlet Allocation and Word2Vec pre-trained on the French Wikipedia corpus. The efficient use of LDA implies the correct setting of the number of topics (K) and not capturing links between topics. The categorization is too broad if the k is low, missing the needed granularity to find weak signals. If the K is high, the words may be "over" categorized, leading to a loss of information due to high precision. Authors vary the number of clusters K to obtain a set of partitions linked together in the form of a tree structure. The method proposes two optimization procedures for finding the optimal K for the LDA, and get the optimal clusters from different cluster levels. The technique presents two indicators :

Remark 1 **Intra-cluster criteria** allow measuring the similarity within a cluster (similarity criterion) based on Word2Vec. The higher the similarity indicator (4), the closer is the intracluster similarity.

$$I_1 = \sum_{w \in E} w2v \text{ Sim}(w_i, w_j) \quad (4)$$

where, I_1 represents the sum of the similarity values of all combinations of word pairs in each cluster, $w2v \text{ Sim}$ is the similarity measure defined in Word2Vec which is the cosine similarity between two vectors.

Remark 2 **Inter-clusters criteria** evaluate the similarity between clusters (consistency criterion). The likeness indicator, Bhattacharyya Distance [89], allows identifying the most pertinent clusters, more likely to represent a weak signal.

The method's evaluation shows the robustness compared to each LDA with $k = 2, \dots, 8$. The work shows the interest in the jointly use LDA and Word2Vec approach.

The primary interest in using LDA is inferring the themes from a corpus of documents. This model has the advantage of proposing a method for detecting weak signals based on tree-multi-clustering, unlike other works that are essentially based on the portfolio maps method. [81] and [44], did not consider the temporal aspect; it would be interesting to study the evolution of a cluster during a period of time and analyze the evolution of a cluster for different cycles of time (e.g., daily, weekly, monthly).

5.2 Graph Theory

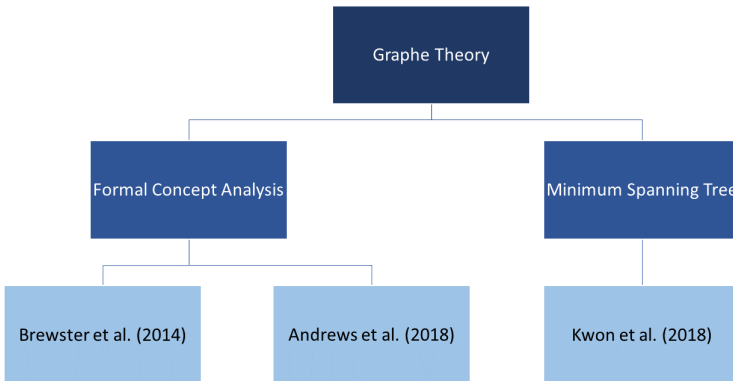


Fig. 11 Taxonomy for the works on graph theory methods

In most recent studies, weak signals have been detected with graph theory way. Those papers ([31; 30 and 63]) explore how graph theory impact weak signal detection

5.2.1 Formal Concept Analysis

[30] and [31] present an approach for early warning in predicting organized crimes. In concrete terms, this technique supports decision-makers in fighting against current and emergent organized crime threats. These researches form part of early Pursuit against Organized crime using environmental scanning, the Law and IntelligenCE systems Project (ePOOLICE). For the rest of the review, [31] propose an approach illustrated using human trafficking and modern slavery as an example.

The authors question the need for assessing and predicting the potential for future threats with weak signals. They allude to "the presence and/or emergence of criminality in citizen-generated content" [90]. For them, a weak signal is intuitively perceived as "a creative inference from a piece of data that is assumed to suggest a link, a potentially meaningful relation between emerging ideas" [18]. Intuitively, the authors consider a weak signal as a "little tangible value (...) isolated indicators" that has "potential for them to be symptomatic of a variety of phenomena" the Canadian Criminal Intelligence Service's (CISC) definitions. In this way, the authors consider the influence of parameters, for example, temporal or geographic proximity to a specific location and type of activity. The mode of data pipeline used by [30] and [31] is similar. ePOOLICE Project includes an environmental crawling via the open-source scanning application on social media, structured and unstructured OSINT repositories.

In the use case of [30] and [31], experts know what they seek, they establish a taxonomy with environmental scanning method PESTLE (91;92) and United Nations Office on Drugs and Crime (UNODC) model [//www.unodc.org/unodc/fr/index.html](http://www.unodc.org/unodc/fr/index.html). Both works use the Formal Concept Analysis method (FCA) to organize information and search for weak signals for the analysis step.

FCA is a mathematical method for implying a relationship between elements [93]. The FCA approach has a number of attractive features: discernment of hierarchy between objects, relationship identification and classification. This structured approach allows visualizing in real-time the concept hierarchies and finds relations and patterns between objects.

It has been suggested in the researches of [94] and [95] that FCA allows identifying atypical behaviors, which is an innovative approach in looking for signals. Moreover, the case studies proposed by [95] and by [94] demonstrate an additional advantage of FCA; identifying atypical behaviors dynamically. Even if FCA presents a series of advantages, it also has a critical limitation: the human intervention required in under-understanding or over-understanding situations. The first is caused by no knowledge, or insufficient knowledge, resulting in false positives. The second is the result of over-knowledge of the field and the parameters' influence, which may not let new trends emerge. Any

element cannot hold certain properties, and this increases noise. If needed, the person who guides the process must know the domain knowledge to determine the objects and objects' attributes. Another limitation of this method is related to the choice of features and properties. These will influence the interpretation, and the wrong set of properties may lead to a false interpretation. FCA is also computationally intensive, as it presents a polynomial computational complexity [96].

5.2.2 Minimum Spanning Tree

[63] use Minimum Spanning Tree (MST) to detect weak signals regarding global listed companies' product and service information. They collect data from ORBIS database (<https://orbis.bvdinfo.com>). The authors compute the co-occurrence matrix between phrases (keywords) which represent products & services. The analysis step is based on the keyword networks analysis and MST graph analysis. The method betweenness centrality over MST shows the importance of a node's mediating role in the whole network. In consequence, a weak signal appears as the smallest node. MST allows visualizing the edges, which minimize the weighted sum of all nodes on a network. The use of betweenness centrality to MST is fully justified. It will show the influence of the node on the network or the diffusion of the event. The proposed methodology is advantageous as detecting weak signals is less time-consuming and complex than it is for other methods.

Their approach combines different techniques to extract the outliers so that each technique's strengths are utilized to identify weak signals. More precisely, by calculating the inverse values, the keywords with low frequency are highlighted and are candidates for weak signals. This study attempted to differentiate node sizes and express them in the MST in the visualization. Then, the use of the betweenness centrality and MST provide an attractive visualization tool. Their study provides excellent results in their fields, but it would be interesting to use other information of business-like patent information, information on products and services of unlisted companies.

5.3 Machine Learning approach

This section presents methods such as Bayesian networks-based approach, clustering-based approach and gradient descent-based approach as shown in Fig. 12.

5.3.1 Bayesian networks-based approach

[34] present a system called NEST (New and Emerging Signals of Trends), which uses clustering, pattern recognition, and cross-impact analysis using a Bayesian network. The goal is to detect weak signals to present supplementary information for future technology researchers. The system is based on Hiltunen's concept [20] where the strength of visibility and the significance of the impact of a particular topic increases over time.

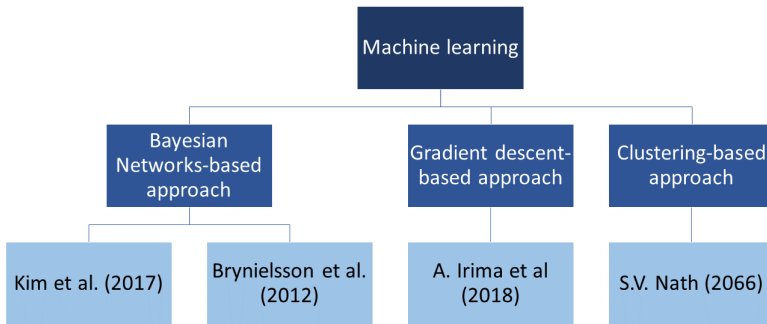


Fig. 12 Works on machine learning approach

The earliest events of a given topic, that become stronger are its weak signals. [34] compute the visibility of a signal with a conceptual evaluation index and cross-impact estimation. The diffusion of an event is computable through a Knowledge-Matrix. The proposed method uses a Bayesian network to quantify the interpretation of the results. Each document is considered a signal in the approach, and it is subdivided into a set of events. Detecting a weak signal means investigating the events of a document cluster and detecting the emerging trends. An emerging trend is characterized by the rise of the frequency of occurrence, the strength of visibility, and the significance of a specific topic's impact. In other words, this network-based trend detection model-related signals are connected. By tracing back the sub-network representing the future trends, weak signals can be found.

NEST has four different goals:

1. creating a systematic process for the identification of weak signals
2. building a reference system supporting futuristic studies (e.g. technology researchers and decision-makers)
3. dealing with unstructured data (for instance, mass media, news, conferences, workshops, academic papers, the Internet) and
4. employing quantitative methods as well as qualitative methods

The approach comprises four steps: Global Monitoring, NEST-clipping, NEST-signal detection and Trend detection. Unlike other proposals, NEST data collection is based on a group of experts called Global Trend Briefing (GTB -[97]). The group of experts propose documents, check the correctness, timeliness, non-duplicability and correctness of the information. This process guarantees a certain quality and coherence over the evaluated signals. The Nest-clipping consists of finding similar information by building a Knowledge Matrix with co-word and clustering them. Each cluster is considered as one topic and it is named manually. Then, the NEST-signal phase concerns detecting and analysing of the topics trying to identify topics with a weak signal

tracking board. The number of generated maps is linked to the number of clusters, which means the method gets more complex with the growth of the number of clusters and their size.

One of the main strengths of NEST is the mechanism put in place to guarantee the quality of the created dataset. Standard procedures are put in place to collect data from different expert sources, reduce noise and ensure the non-duplication of collected information, which is quite interesting. Even if the collection of documents relies on the referral of experts, which is a potential source of bias. The system can only observe what the experts consider potentially relevant. However, specialists often overlook weak signals [56], and if the collection of topics is not broad enough, the system may be blind to some critical future changes.

[78], [98] propose two different ways to detect weak signals: a weighted average model or Bayesian belief networks. The use of weighted value requires a predetermined weight that expert assigns. Each parameter owns a degree of importance. This method is interesting, but the expert's intervention could lead to a distorted assessment. The authors objective is to determine, from posts on web forums, the interest of different users (alias) over the same topic. However, the way authors propose the application of Bayesian networks would imply the creation of one network for each alias, which could be hard to implement and handle for large forums.

5.3.2 Gradient descent-based approach

[80] apply machine learning techniques to identify weak signals. Gradient Descent is a machine learning technique where a convex function is used to verify the progress of a method as the parameters are interactively changed. The lower the value, the best is the result, as we want to minimize the objective function. This idea is used in, to find weak signals through the use of supervised learning. For the authors, weak signals' detection usually relies largely on tacit human knowledge. Their study focuses on weak signals that are "the information that a human reader is able to extract from a piece of news which is no more than a hint that a certain event is going to happen". The used dataset is composed of 40,000 scientific articles published in the last 50 years. For the data analysis step, a team of people annotated each piece of news binarily, and they selected the documents where the whole team agreed. The second round of annotation restricts the scope to paragraphs (containing 100-250 words). In this phase, they evaluate the presence of weak signal corpus for short term (referring to paragraph) and long term (referring to full documents). Then, they provide an evolution of the average number of documents on which there was a strong disagreement.

As mentioned by the authors, the use of TFI-DF provides negligible accuracy. According to them, researching a weak signal with tf-idf is not only sufficient. In order to overcome this problem, they will use a couple of

approaches: QDA (scikit library ²) & SVM (Weka library ³) and gradient descent. Their study highlights that machine learning methods can be used efficiently for tasks where tacit human knowledge plays an important role, and we can create annotated datasets.

5.3.3 Clustering-based approach

[77] proposes a clustering-based method with two objectives, the first detecting crime patterns and the second improving the productivity of detectives and other law enforcement officers. The main challenge faced in crime analysis is treating fields such as the modus operandis, a free text field. Finding ways to structure these fields so that regular methods can treat them is not an easy task. The proposed method consists of four steps, (1) collecting data information (public information, over sheriff's office sites), (2) defining the clusters, (3) clustering the collected information and (4) identifying. A cluster is a group of crimes in a geographical region in the presented work.

The clustering method explored is k-means. Attributes are dynamically weighted and clustered regarding the different types of crimes. Thanks to the clusters, it is possible to represent them using geospatial plots in a map (describing the hotspots of crime). The detective can then analyze the crime patterns. Using k-means and the scheme for weighting the significant attributes convey decision-making. However, the use of K-means requires setting a K-number of clusters and so to know approximately input data and the number of classes available. Also, the use of K-means implies that the initial k cluster seeds may profoundly impact on the distribution of the classes.

5.4 Semantic approaches

Previous attempts to scan weak signals from quantitative data focus on earliness but neglect signals' evolving nature.[33] and, [78] and [98] propose to take into account the semantic as shown in Fig. 13. [33] considers the semantic of signals during their evolution through time.

In their work, authors use Latent Semantic Indexing (LSI) [99] based on Singular Value Decomposition (SVD) to create a cluster of documents. Then they investigate these clusters and measure their evolution at successive points. Authors focus on the development of the discourse, as different people use different writing styles to describe the same event. The objective is to understand the evolution of signals through the evolution of the semantic textual patterns of the identified clusters. With this, they intend to evaluate if a signal will stay weak or may become strong. For the data collection, [33] study environmental scanning by crawling different websites in the German language at different points in time. LSI considers the term dependencies to identify semantic generalization. Notice that the number of clusters is found empirically ($k = 25$).

²<https://scikit-learn.org/0.16/modules/generated/sklearn.qda.QDA.html>

³https://waikato.github.io/weka-wiki/use-weka_in_your_java_code/

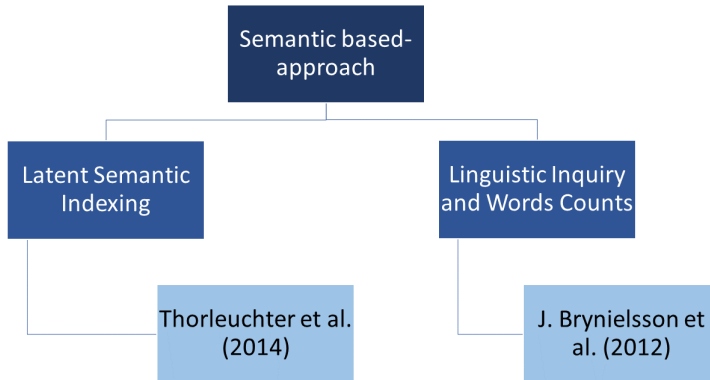


Fig. 13 Works on semantic based-approach

By considering the semantic of the words,[33] address the problems of polysemy and homonyms of traditional lexical matching. Even though computationally intensive, SVD enables the calculation of the most significant number of relevant weak signals. Using topic modeling techniques to detect a cluster of documents is interesting; however, the authors do not present any study over the optimal number of clusters for a given topic. The $k = 25$ is an empirical value. If k is too small, the clusters will be too broad, and If k is too large, groups will capture what could be considered noise at first. The detection of weak signals would require the overlapping of different clusters.

[78], [98] propose a method to collect information about people that are on the verge of preparing and committing lone violent acts (i.e. lone wolf). Their approach involves breaking down this complex problem into sub-problems and fusing the results into a global solution. They consider terrorist potential like a weak signal and establish relations between aliases (usernames employed in websites) to find authors that use different aliases. The work's challenge is determining the degree that an individual is/will become a lone wolf attack. Their definition of weak signal is related to the context and the field of knowledge. Naturally, a weak signal is a subtle key (an alias) that has a significant impact. In this study, the authors propose a semantic analysis (WordNet ⁴) to analyze individuals' speech. However, this method does not consider the sentence thone (ironic, jokes,...), which may lead to false positive.

Authors gather data by using a crawler and hyperlink analysis. From web page URLs of well-known terrorist web-sites, it is possible to follow the source page hyperlinks to find more web pages [100]. These results provide an extensive network graph, where the nodes are the web pages and the edges, the hyperlink found. To extract relevant websites, the authors suggest using topic-filtered web harvesting. All aliases are extracted, and a model is created for each alias from this organized text. In this study proposition, each alias receives

⁴<https://wordnet.princeton.edu/>

a score regarding the presence of hate, or violent comments [101], which affects the analysis. A alias model is composed of all relevant information related to the alias. For the analysis, using these indicators requires the classification of radical and non-radical content, semantic relations (lexical database WordNet), the identification of similarities (Linguistic Inquiry and Word Count - LIWC ⁵). [78] assume that when an author makes a post on a discussion board, this reveals his interest in the topic, which can be used to identify aliases. The evaluation of the alias is closer to the event's diffusion (i.e. issue axis of Hiltunen's concept). This task could make it possible to quantify the importance of an alias or the diffusion of its actions on the network. Two methods are proposed, a weighted average model and a Bayesian belief network. The weighted average model requires experts to provide the weights. There is a risk of adding a cognitive bias, as the weights imply the importance of the parameters. The Bayesian network is immune to this bias, as they may be created without the expert's intervention, but the complexity is higher, as a new network is required for each alias.

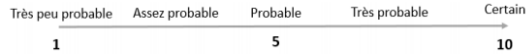
$$I(x) = f(\{J(w) : w \in W \wedge A(x, w) = 1\}, C(x)) \quad (5)$$

where I in (5) is the result of the interestingness function over each alias x , J the interestingness function for a web site, $C(x)$ a content analysis function and $A(x, w)$ is equal to 1 if x is active. The content of posts may reveal which alias is behind that specific post, compared to other aliases. A step of matching is realized to identify an alias that has several aliases from the ranking. Different methods are proposed: Jaro-Winkler metric [102], Levenshtein list of potential extremists [98]. The work does not present results to evaluate the efficiency of the proposed methods. However, the concepts and ideas are interesting, particularly using techniques such as topic-filtered web harvesting and content analysis using natural language processing. Moreover, the recognition of the document author through the content analysis is another interesting contribution.

5.5 Expert-based approach

[79] focus on organizing the information and on the identifying of weak signals to enable efficient innovation for companies. For the authors, a weak signal is a piece of information that does not expose an immediate interest in the first view. However, in a given context, they may represent a fundamental piece of information [36]. To find them past and present need to be evaluated [103]. They propose the following method, (1) fixing a subject of research, (2) defining a source of signs, (3) gathering data, (4) classing with keywords or areas of activities and interpreting the results. Fixing a research subject means map topics with X-Mind and (5) identifying relevant opportunities and selecting the relevant topics. The identification step is based on the principle of scoring (Eq.(6)) with is the likelihood of occurrence of the theme in the company

⁵<https://liwc.wpengine.com/>

**Fig. 14** Evaluation of occurrence probability**Fig. 15** Evaluation of the impact

noted P in Fig. 14 and the impact on the business, noted I in Fig. 15. P is a likelihood assessment with the scale and I the scale of impact.

The X-Mind maps are then ranked by experts that assign priority, "not urgent", "to treat secondarily", or "to treat as a priority". Then, analysts can plan their actions according to the score of importance. The idea of assigning a priority level is interesting but relies on the expertise and evaluation of a group of experts. The result is mainly based on the experts' intuition, experience and commitment, as the work may be long and tedious. We can expect the final result to provide cognitive bias and loss of performance from the expert.

The score of Importance is :

$$Score = I \times P \quad (6)$$

6 Conclusions

This survey reviewed different methods for the detection of early warning signals. Early warning signals can be used in a relatively large range of situations, ranging from anomaly to criminal detection, including decision-making and anticipation in the context of companies' strategic planning. This survey discusses the difficulties of the field and the ground principles on which researchers rely on. In Section 4 presents how different methods fit a standard data treatment pipeline and in Section 5 a categorization of the available methods. The proposed taxonomy divides the reviewed methods into: statistics-based, Bayesian networks-based, semantic-based, gradient descent-based, clustering-based and expert-based.

The research in weak signals is largely influenced by Ansoff's initial works in the 70's and Hiltunen's concepts in 00's. Hiltunen's works introduce the importance of a signal and the graduation of a signal into weak to strong one. Some recent studies extended this concept using the portfolio maps method. Conversely, [64] instead focused on the signal rarity and its discordance to the context. Kim's work highlights the importance of the complementarity of data sources by merging two relevant databases. For [29] the changes in appearance rate of keywords may reveal the evolution of weak signals into strong ones. Some works try to take into account temporal aspects to track the evolution signals ([33; 46; 47]), even if we still have room for works in this field. Reliable

methods for evaluating the probability of a signal to transform from weak to strong are fundamental in forecasting.

The growth in the volume, kind, and quality of available data drives the development of innovative approaches to dealing with it. Companies, governmental agencies, and politics, among others, can get a real-time perspective of what has the potential to become important through sources such as social networks, where people can freely express themselves. This real-time, information flow and the possibility of detecting early signals of changes may represent a considerable advantage for different actors. Weak signal research is an active topic, and we hope to see even more advancements in the future, considering all of the potential interests.

Appendix A Section title of first appendix

Table A1: Weak signal definition

References	Quotes
[5]	"Early indications about impactful events (...) all that is known (of them) is that some Threats and Opportunities will undoubtedly arise, but their shape and nature and source are not yet known"
[15]	" sudden, urgent, unfamiliar changes in the firm's perspective which threaten either a major profit reversal or loss of a major opportunity "
[27]	"Weak signals are first symptoms of strategic discontinuities; they are symptoms of possible change in the future"
[104]	"seemingly innocuous 'data' but with an interpretation that can trigger an alert. This alert indicates that an event may occur that could have a significant impact (in terms of opportunities or threats). After interpreting, the signal is no longer qualified as a weak signal, but as an early warning signal"
[18]	"Gross, unstructured, fragmented, incomplete and inadvertent environmental data that may be refined into valuable information regarding context and further be articulated into strategically actionable knowledge" and "premature, imperfect information (...) obfuscated by confounding factors "
[105]	" as information on potential change of a system to an unknown direction."
[106]	"the early signs of possible but not confirmed changes that may later become more significant indicators of critical forces for development, (...) the first signs of paradigm shifts, or future trends, drivers or discontinuities"
Continued on next page	

Table A1 – continued from previous page

References	Quotes
[78]	"the relevant clues"
[19]	"An idea or trend that will affect how we do business, what business we do, and the environment in which we will work"
[16]	"Hardly perceivable at present but constitutes a strong trend in the future"
[39]	"Current oddities, strange issues that are thought to be in key position in anticipating future changes in organizational environments"
[29]	"Imprecise and early indicators of impending important events or trends, which are considered key to formulating new potential business items"
[64]	"they are anomalies or strange issues that are not compatible with the prevailing sensemaking paradigm (...). novel future signals (documents or keywords), in futuristic data that are not only rare but also have paradigm unrelatedness"
[46]	"imprecise and non-mainstream signals with a potential to change in the future"
[43]	" These indicators of change can be advanced, somewhat noisy, and generally socially situated trends and systems that constitute raw information to enable anticipatory action. (...) represent an unknown, unexpected, or rare change, which makes them hard to distinguish as relevant."

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